

Lab 3: Inverting Amplifiers and Weighted Summer

ECE 0102: Microelectronic Circuits + Lab

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Team Roles:

Activity	Student Name
Circuit Construction	Stefan Zhang
Data Collection	Knight Liu
Data Analysis	Stefan Zhang; Knight Liu
Answers to Questions	Stefan Zhang; Knight Liu
Lab Report Writing	Stefan Zhang

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1) Objectives and Background

The objective of this lab is to study the operation of inverting amplifiers and to design a weighted summing circuit using operational amplifiers. The measured results are compared with theoretical predictions to evaluate circuit performance.

2) Equipment Used

Table 1: Equipment used in Lab 3

Item	Quantity
LM741 Operational Amplifier [3]	2
Resistors	Several
Function Generator	1
Oscilloscope	1
DC Power Supply (+12V / -12V)	1
Breadboard and wires	1 set

3) Action 1.1: Inverting Amplifier with $V_{REF} = 0$

3.1) Theory

For the circuit in Fig. 5 of the lab manual, the general output expression is [1, 2]:

$$V_{OUT} = -\frac{R_2}{R_1}V_{IN} + \left(1 + \frac{R_2}{R_1}\right)V_{REF}$$

In Action 1.1, $V_{REF} = 0$, so the equation becomes:

$$V_{OUT} = -\frac{R_2}{R_1}V_{IN}$$

The potentiometer was adjusted until the gain was approximately -3 , which means:

$$-\frac{R_2}{R_1} \approx -3$$

Since $R_1 = 10 \text{ k}\Omega$, the expected value of R_2 is approximately:

$$R_2 \approx 30 \text{ k}\Omega$$

3.2) Experimental Setup

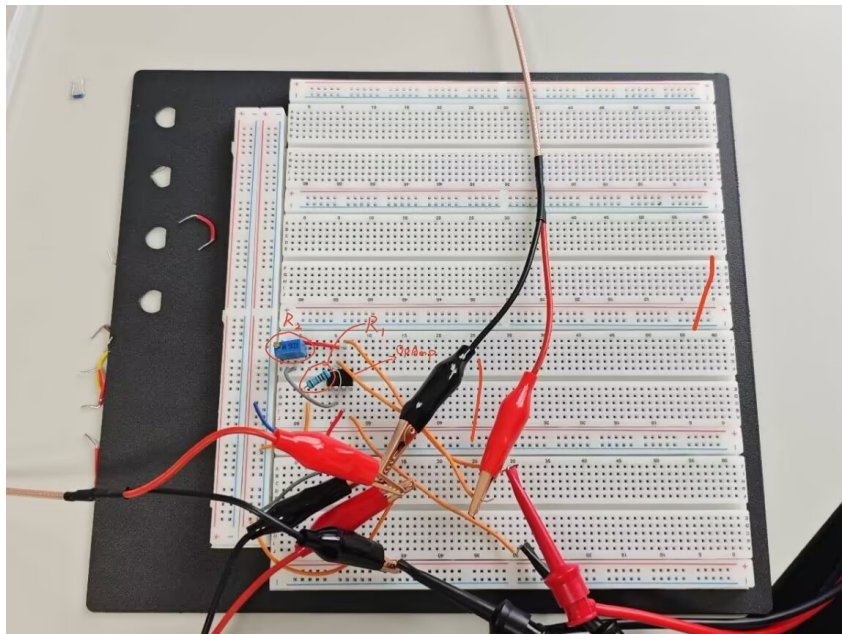


Figure 1: Experimental setup for Action 1.1.

3.3) Experimental Results

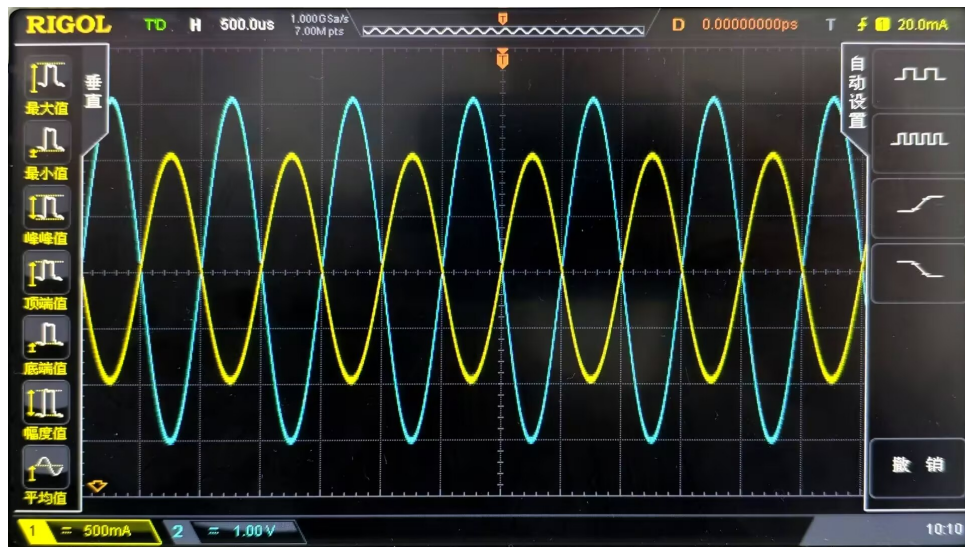


Figure 2: Measured V_{IN} and V_{OUT} waveforms

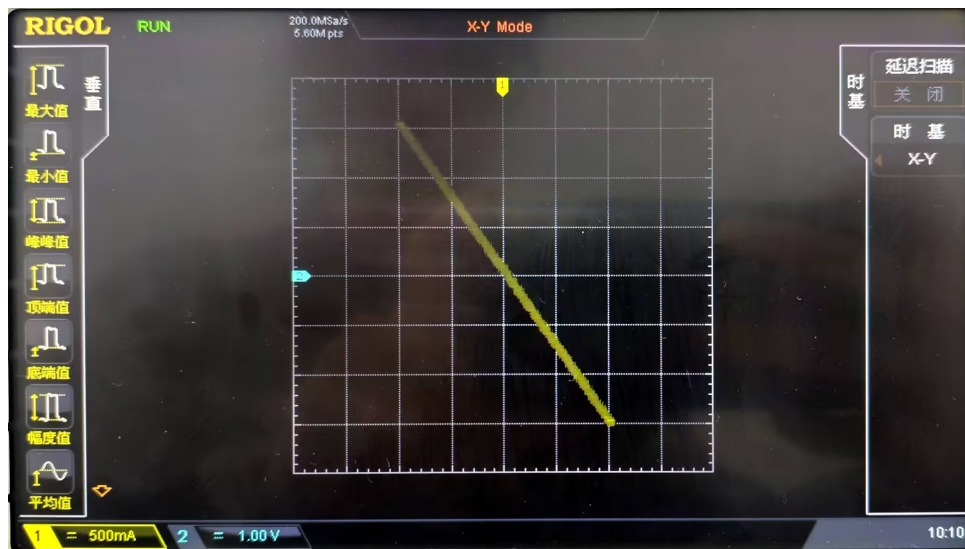


Figure 3: Transfer characteristics (X-Y mode)

3.4) Discussion

The output is inverted and amplified with a gain close to -3. The waveform agrees with theoretical prediction with small deviation due to non-ideal op-amp behavior.

3.5) Measured Value of R_2

After disconnecting the potentiometer from the circuit, the measured resistance of R_2 was approximately 30.019 k Ω .

This value is close to the expected resistance because the gain of an inverting amplifier is:

$$A_v = -\frac{R_2}{R_1}$$

To obtain a gain of approximately -3 with $R_1 = 10 \text{ k}\Omega$, the required resistance is:

$$R_2 \approx 3R_1 = 30 \text{ k}\Omega$$

Any small difference between the measured value and the theoretical value can be caused by potentiometer adjustment error, meter reading uncertainty, and the fact that the gain was tuned experimentally rather than set to an exact resistance value.

4) Action 1.2: Inverting Amplifier with $V_{REF} = 1V$

4.1) Theory

$$V_{OUT} = -\frac{R_2}{R_1}V_{IN} + V_{REF} \left(1 + \frac{R_2}{R_1}\right)$$

4.2) Experimental Setup

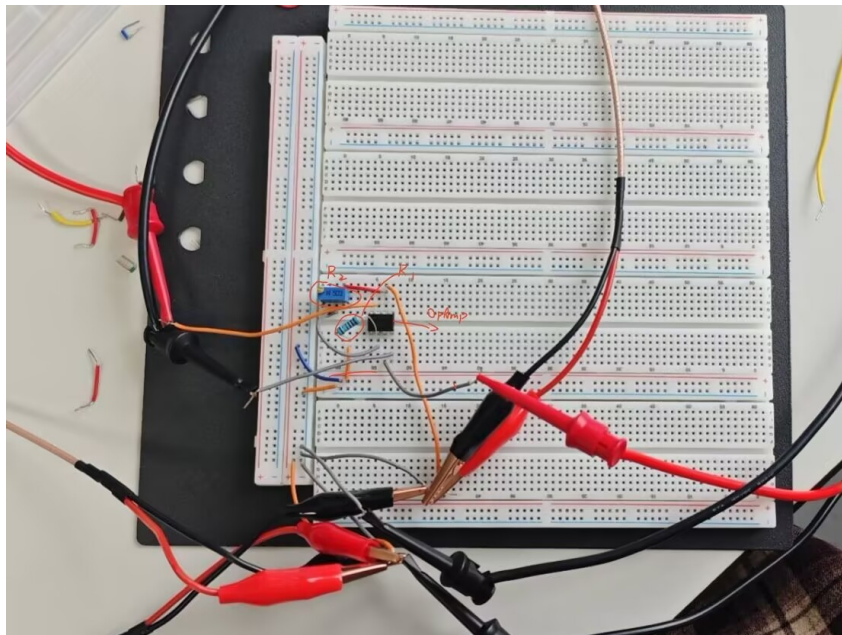


Figure 4: Experimental setup for Action 1.2.

4.3) Experimental Results



Figure 5: Measured waveforms with $V_{REF} = 1V$



Figure 6: Transfer characteristics

4.4) Discussion

Compared with Action 1.1, the waveform in Action 1.2 is vertically shifted because the nonzero reference voltage adds a DC offset to the output. The measured slope in the X-Y display remains nearly the same, which shows that the gain is still approximately -3 . The small deviation from the ideal result is mainly caused by resistor tolerance, offset voltage, and other non-ideal characteristics of the LM741 op-amp.

5) Action 2.1: Weighted Summer Design

5.1) Theory

The required equation is:

$$V_O = v_1 + 3v_2 - 2v_3 - 5v_4$$

A single inverting summing amplifier would reverse the sign of all input terms, so two op-amp stages were used.

In the first stage, an inverting summer generates:

$$V_X = -(v_1 + 3v_2)$$

To obtain this relation with $R_{f1} = 10 \text{ k}\Omega$, the resistor values are chosen as:

$$R_{v1} = 10 \text{ k}\Omega, \quad R_{v2} = 3.33 \text{ k}\Omega$$

In the second stage, another inverting summer combines V_X , v_3 , and v_4 :

$$V_O = -V_X - 2v_3 - 5v_4$$

Substituting V_X gives:

$$V_O = -[-(v_1 + 3v_2)] - 2v_3 - 5v_4$$

$$V_O = v_1 + 3v_2 - 2v_3 - 5v_4$$

For the second stage, with $R_{f2} = 10 \text{ k}\Omega$, the resistor values are:

$$R_{VX} = 10 \text{ k}\Omega, \quad R_{v3} = 5 \text{ k}\Omega, \quad R_{v4} = 2 \text{ k}\Omega$$

Thus, the required weighted summing function is achieved using two LM741 op-amps.

5.2) Circuit

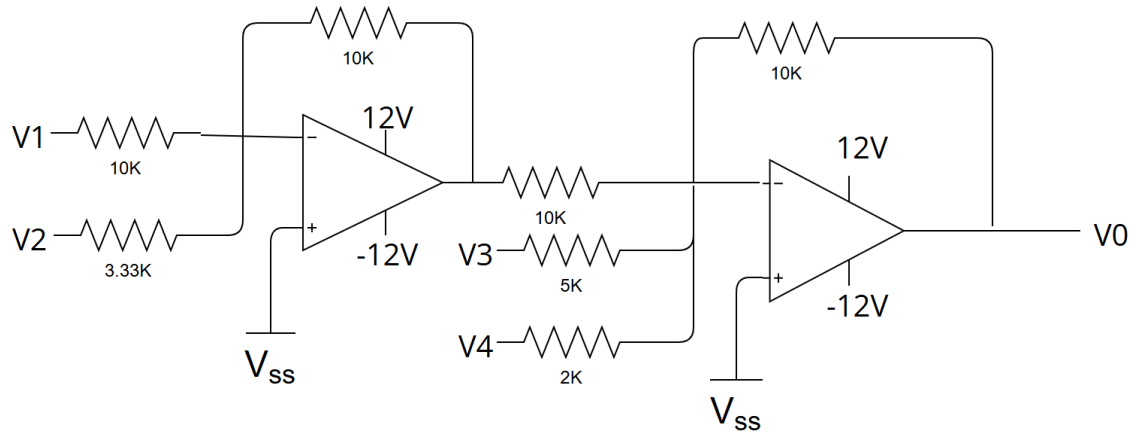


Figure 7: Weighted summer circuit schematic

6) Action 2.2: Experimental Results

Table 2: Measured and calculated results

V1	V2	V3	V4	Measured V_O [V]	Calculated V_O [V]	Error [%]
0	0	0	0	0.022	0	N/A
1	0	0	0	1.165	1	16.5
-1	0	0	0	-0.869	-1	13.1
0	1	0	0	2.964	3	1.2
0	-1	0	0	-2.984	-3	0.5
0	0	1	0	-1.967	-2	1.6
0	0	-1	0	1.996	2	0.2
0	0	0	1	-5.050	-5	1.0
0	0	0	-1	5.200	5	4.0
1	1	1	1	-2.961	-3	1.3
-1	-1	-1	-1	3.151	3	5.0

6.1) Discussion

The measured results are close to theoretical values. Larger errors appear when the output magnitude is small or when multiple inputs are combined. The main sources of

error include resistor tolerance and op-amp non-idealities.

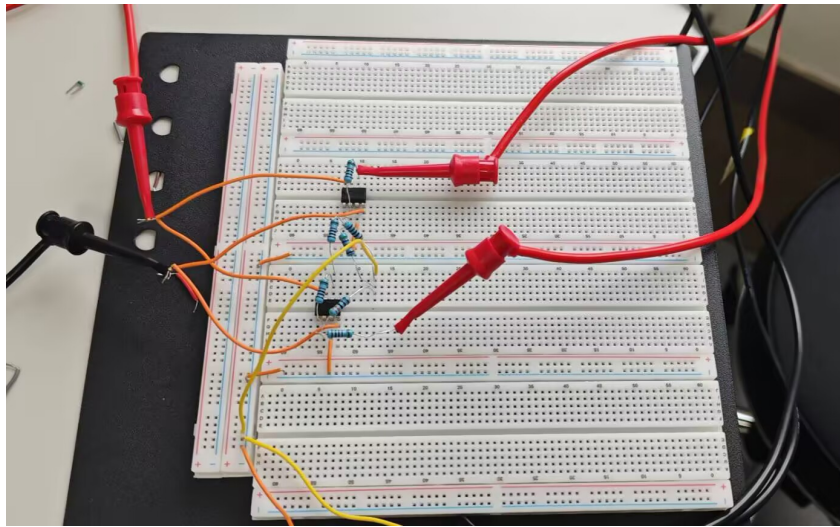


Figure 8: Constructed weighted summer circuit

7) Conclusion

In this lab, inverting amplifiers were analyzed and tested. The designed weighted summer successfully implemented the required function. The experimental results generally match the theoretical predictions, with small deviations due to practical limitations.

References

- [1] SCUPI 2025S Microelectronics Circuit Lab Manual, Lab #3: Inverting Amplifiers and Weighted Summer.
- [2] Sedra, A. S., Smith, K. C., *Microelectronic Circuits*, Oxford University Press.
- [3] LM741 Datasheet, Texas Instruments.